

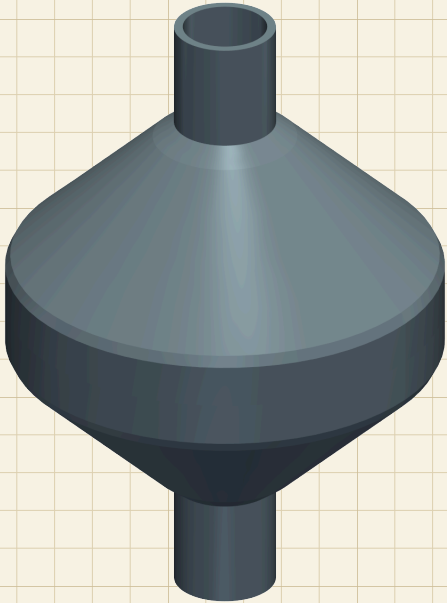
F100 / WATER FILTER

Seven components. One pressure-boundary weld.

DESIGN INTENT

A flat, directly contacting coarse / fine / coarse pack, captured between two annular frames and two identical machined cups. The outer orbital weld contains pressure; the coined frames and closed element seam prevent internal bypass. Service basis: water at 70 °F; 650 psi inside relative to atmosphere. Pressure ratings do not apply across a blocked screen. ^[15]

F100 / ASSEMBLED ISOMETRIC



Build123d geometry, transparent VTK rendering. Ø82 mm body; 116 mm overall.

CONCEPT SCREENING

Item	Result	Meaning
Dry mass	0.802 lb	Calculated; 20% growth still below 1 lb
Clean Δp, nominal / surge	0.48 / 3.83 psi	Conditional hydraulic hypotheses
Housing at 1300 psi	~199 MPa local elastic peak	Below 275 MPa yield; peak not converged
99.9% at 3 μm	Not substantiated	Fine media selection is the first gate
Crush / bypass seal	Development required	Geometry alone cannot certify sealing

REVIEW POSITION

This is a detailed development candidate, not a qualified pressure component. The strongest evidence is the mass model, exact seven-solid assembly, and elastic housing screening. No supplier has yet substantiated all retention, pressure-drop and dirt-capacity requirements for the selected fine-cloth envelope.

01 / REQUIREMENTS AND INTERPRETATION

This section lists what the filter must do and how each requirement is being checked.

The original pressure values were written as psid. The subsequent clarification governs: they are internal-to-external housing pressure differences, conventionally psig here. Pressure-drop limits are inlet-to-outlet differentials, reported in psi. The 5 psi clean limit is conservatively applied at both nominal and surge flow. The 18 psi loaded limit is evaluated at nominal flow, with surge also reported. Contaminant type and surge duration remain unspecified.

REQUIREMENT-BY-REQUIREMENT ASSESSMENT

ID / requirement	Evaluation	Closure
R01 / MEOP 650 psig	Elastic housing screen favorable	Qualified weld + leak/pressure test
R02 / proof 1300 psig	Linear peak ~199 MPa; no test	Hydrostatic proof; no permanent set
R03 / burst ≥ 2600 psig	Gross wall promising; local screen unresolved	Nonlinear analysis + destructive test
R04 / 5 μm rating	5-S geometric rating; GKD alternative	Define absolute criterion; particle challenge
R05 / 99.9% at 3 μm	$\beta_3 \geq 1000$; no qualified catalog match	Lot and completed-assembly challenge
R06 / 0.14 lbm/s	3.818 L/min at 70 °F	Measured assembly flow curve
R07 / surge 0.47 lbm/s	12.817 L/min; duration unknown	Flow transient, shedding and retention
R08 / clean $\Delta p \leq 5$ psi	0.48 nominal / 3.83 surge, assumed Cv,Ci	Supplier coefficients + clean-flow test
R09 / dirt ≥ 0.318 g	Conditional on cake and contaminant	Specified retained dry dirt loading test
R10 / loaded $\Delta p \leq 18$ psi	0.90 nominal for $\alpha = 10^9$ m/kg	Measured dirty-flow / efficiency curve
R11 / mass ≤ 1 lb	0.802 lb dry; 0.962 lb with 20% growth	Weigh actual dry finished filter
R12 / water at 70 °F	21.11 °C; fixed design temperature	Temperature-controlled qualification
R13 / seven components	7 CAD solids; 5-piece element	Drawing and assembly inspection

REMAINING SERVICE DEFINITIONS

Water chemistry, particle material/shape and size calibration, dirt feed concentration, lifetime throughput, pressure-cycle count, surge duration, allowable reverse flow, external line loads and installation line alloy require definition. These are open inputs, not silently satisfied requirements.

02 / ARCHITECTURE AND MATERIAL TRADE

This section explains the seven parts and why each material was chosen.

Select commercially pure titanium Grade 2 for the two housing halves. Use annealed 316L sheet frames and a stainless fine/coarse pack. This hybrid arrangement avoids a titanium-to-stainless fusion joint: the internal pack is mechanically captured. The external stubs are integral titanium and assume compatible titanium lines. Stainless installation lines would require a separately engineered transition, outside this seven-part scope.

SEVEN-PIECE BILL OF MATERIALS

Part	Qty	Nominal definition
F100-01 housing	2	Identical CP Ti Grade 2 cups; integral $\frac{3}{4}$ in stubs
F100-02 frame	2	316L annealed sheet; Ø70 / Ø58 × 0.200 mm blank
F100-03 support	2	40 × 40, Ø0.254 mm wire; 0.480 mm calender target
F100-04 fine cloth	1	5-S envelope, 0.180 mm; performance approval pending

WHY TITANIUM HOUSING

Option	Mass estimate	Decision
Ti Grade 2 cups / SS element	0.364 kg / 0.802 lb	Baseline; material and weld coupons required
Same geometry, all 316L	~0.631 kg / 1.392 lb	Fails 1 lb before growth
All titanium media/frame	Not established	No sourced β3 or welding schedule

MATERIAL BASIS AND INTERFACES

Use 275 MPa minimum yield and 345 MPa minimum tensile as room-temperature screening values for Grade 2, with 105 GPa elastic modulus and assumed Poisson ratio 0.34. Actual pressure-component stock must carry the applicable product-form certificate; cited catalog data do not certify an 82 mm machining blank. ^{[6] [7]}

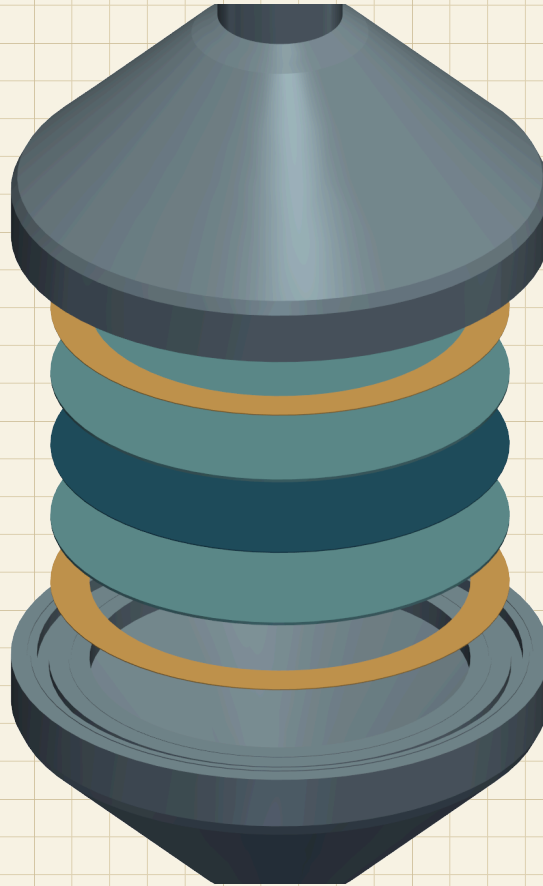
The coarse catalog examples are T316, not certified 316L. The baseline drawing requires a certified 316L equivalent with the same construction and recalendered thickness, or a documented material change after weld development. Keep titanium and stainless fabrication tools segregated. Evaluate galvanic/crevice behavior for actual water chemistry.

03 / ASSEMBLY AND EXPLODED VIEWS

This section shows how the seven parts fit together.

The order is housing / frame / coarse / fine / coarse / frame / housing. The three mesh layers touch across the full active face. Both external frames are continuous annuli. All five element layers participate in the closed perimeter attachment.

SEVEN COMPONENTS / SEPARATED AXIALLY



Same CAD parts as the assembled model. Mesh discs are colored envelope solids, not pore-resolved woven wires.

READING THE CAD

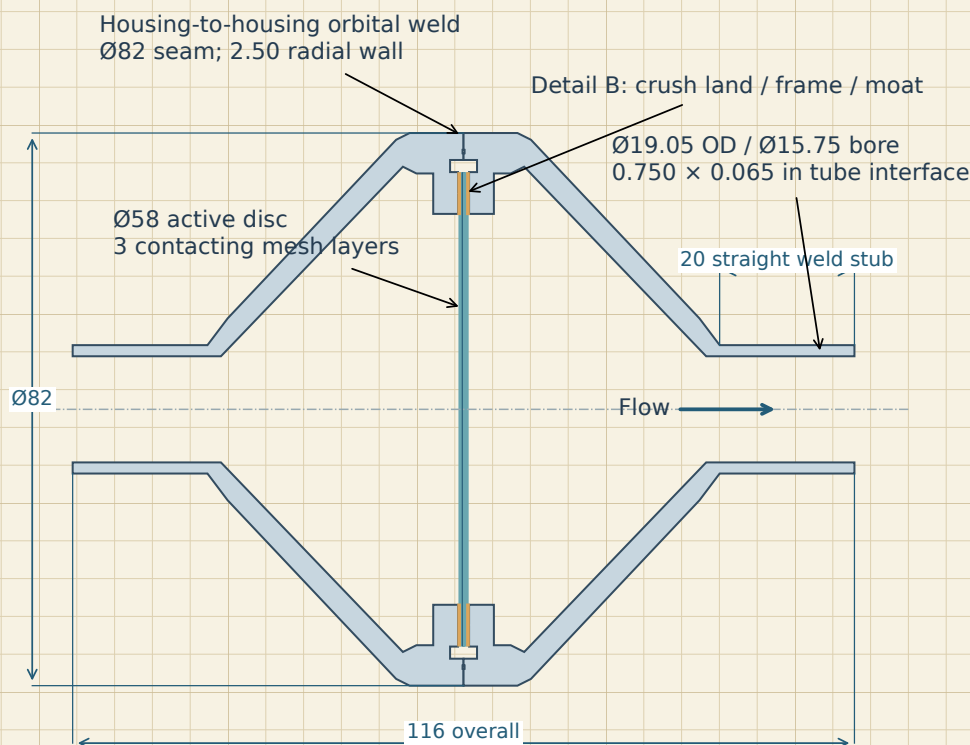
The STEP assembly contains seven valid, noninterfering solids. Fine and coarse discs are deliberately represented by their physical envelopes; treating those envelopes as solid plates in a structural or fluid solver would be incorrect. The colored render does not imply that water passes through solid CAD discs. Frames in the assembled model use a 0.170 mm coined envelope; the separate frame-blank STEP retains 0.200 mm stock thickness.

04 / HOUSING GEOMETRY AND ORBITAL-WELD RATIONALE

This section explains the housing shape and where its outer weld goes.

The axisymmetric cups combine an expanded element chamber with a conical transition to a straight tube stub. A cone carries pressure through membrane action more efficiently than a broad unsupported flat end cap, while providing a smooth flow transition and tool access. The small changes of slope are explicit sharp concept vertices; production blend radii remain a drawing-release task and must be included in the next stress model.

AXIAL SECTION A-A / DIMENSIONS IN MM



Dimensioned profile derived from design.py; symmetric halves about z=0. Section is schematic only in mesh porosity.

WHY THE JOINING RIM IS CIRCULAR

The Ø82 mm rim presents a constant-diameter circumferential square butt joint with a 2.50 mm radial fusion wall. Two identical ends permit repeatable concentric fixturing and a single orbital GTAW seam. The sealing land is at radius 32 mm; the independent closure stop is at radius 37–38 mm; a root relief separates the stop from the fusion wall at radius 38.5–41 mm. This prevents the element from being intentionally consumed in the housing weld.

An orbital head covering 82 mm diameter is available as a family, but its collet envelope and clearance over the short shoulder are not verified. Plan a fixture-specific open head or rotary welding station if a closed head cannot clear the cups. One seam may require multiple passes; “one weld” is not a promise of one pass. ^[9]

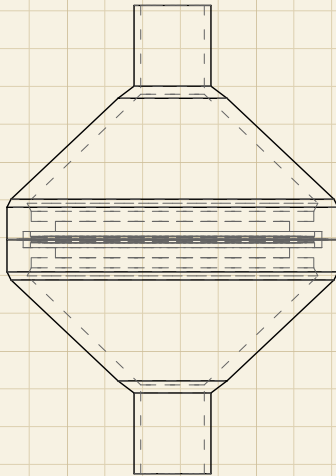
05 / CAD DRAWING SHEET

These drawings show the filter from the outside and sliced through the middle.

Build123d exports the solid assembly, exploded assembly, identical housing half and sheet blank as STEP. The center-plane section is also exported as a 1:1 millimeter DXF. Projection views below are actual OpenCascade visible/hidden-edge views, scaled to fit.

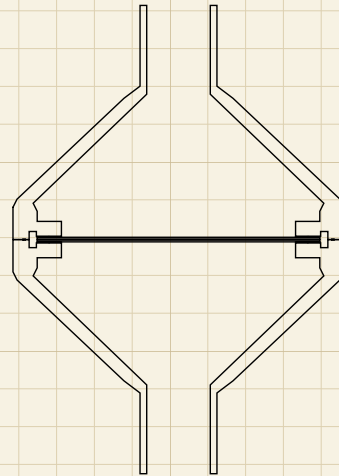
FRONT, END, CENTER SECTION AND HOUSING HALF

FRONT / OUTSIDE



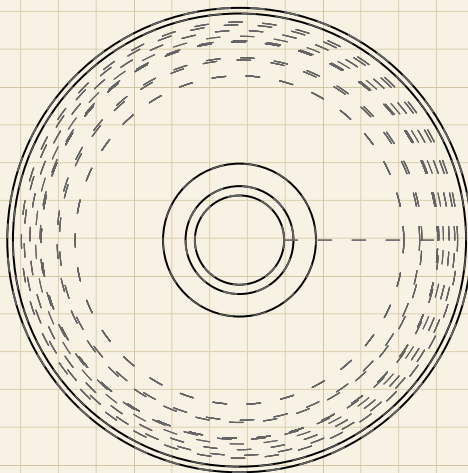
116 mm overall; 82 mm outside diameter

SECTION A-A / THROUGH THE CENTER



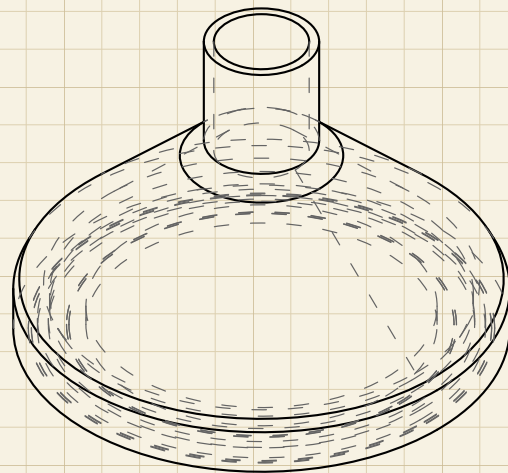
XZ plane, $y = 0$; actual CAD intersection

END / LOOKING INTO THE TUBE



58 mm active face; 70 mm frame

ONE HOUSING HALF / USE TWICE



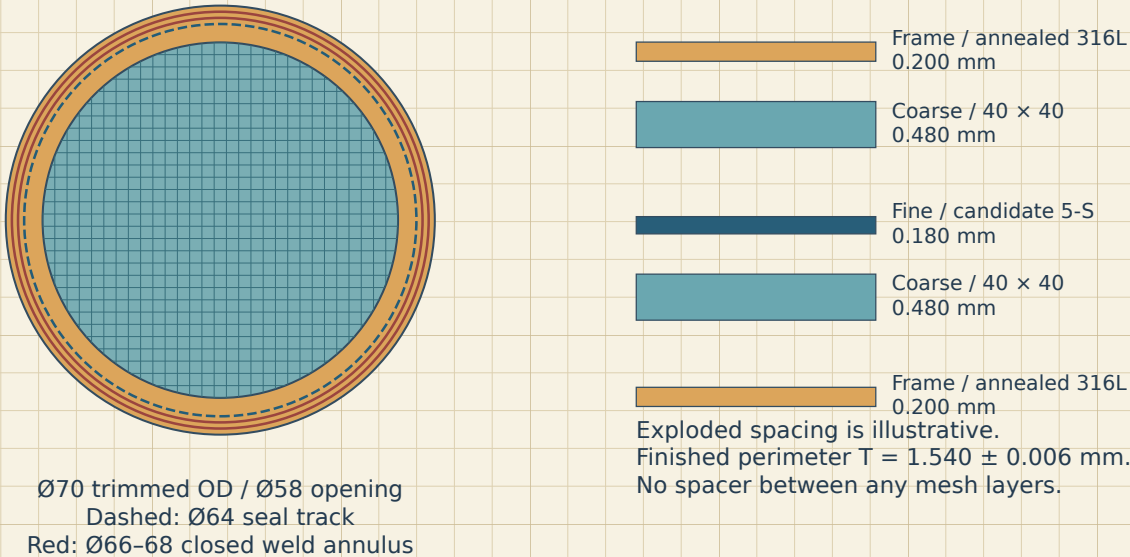
Opposing half is the same part, reversed

Section A-A is a true 2D intersection of the seven-part CAD assembly with the XZ plane at $y=0$. Mesh appears as thin envelopes; individual pores are not modeled. Views are scaled to fit.

06 / FILTER ELEMENT AND SUPPLIER DEFINITION

This section shows the filter layers and identifies suppliers that could provide the mesh. The fine-cloth baseline is a dimensional and hydraulic development candidate, not an approved β_3 element. Haver’s specific 5-S construction provides traceable physical data; GKD provides a second traceable 5 μm weave family. The 3 μm efficiency requirement can force a finer custom cloth and a fresh hydraulic/stack calculation.

FRAMED THREE-MESH PACK / PLAN AND STACK



Python/Matplotlib vector illustration: circles show the annular features and rectangles show the layers. The coarse mesh grid and exploded spacing are illustrative, not supplier-accurate weave geometry. Weld annulus Ø66–68, seal-track mean Ø64, trimmed OD Ø70; mesh remains intact across Ø58.

TRACEABLE MESH SOURCING

Supplier / item	Supported facts	Not established
Haver / RPD HIFLO 5-S	5 μm geometric pore; 0.18 mm	β_3 , Cv/Ci for water, exact alloy/lot
GKD / 5 μm DTW family	5 μm absolute-opening end of family	Exact SKU, thickness, β_3 , permeability
TWP / 040X040T0100	40 × 40; 0.254 wire; T316	316L certificate; calendered performance
Darby / 40316.010PL	Same coarse geometry; T316	316L certificate; lot origin
Dorstener / custom fabrication	Welding, laser trim, cleaning	Exact fine media and qualified seam

SOURCE TRAIL

Fine-cloth references: ^[1] ^[2]. Coarse cloth references: ^[3] ^[4]. Fabrication lead: ^[5]. Require origin of the actual wire, weaving and processing if non-Chinese sourcing is retained; headquarters location is not origin evidence.

07 / RETENTION AND THE BYPASS BUDGET

This section explains how well the filter must catch particles and why water must not leak around the mesh.

Interpret liquid efficiency as number-based capture at and above a defined 3 µm particle size: 99.9% corresponds to $\beta_3 \geq 1000$. A 5 µm geometric pore or “absolute opening” statement does not demonstrate that efficiency at 3 µm. No source located for this report verifies both requirements for a purchasable fine-cloth item. This is the principal open compliance issue.

EFFICIENCY RELATION

$$\beta_3 = \frac{N_{up}}{N_{down}} \quad (1)$$

$$\eta_3 = 1 + \frac{-N_{down}}{N_{up}} \quad (2)$$

MEDIA PLUS ASSEMBLY MUST MEET THE REQUIREMENT

For a bypass flow fraction b carrying feed concentration, assembly efficiency is $\eta_a = (1 - b)\eta_m$. If the cloth itself is only 99.9% efficient, essentially no bypass budget remains. A provisional allocation of 99.95% media efficiency and 0.05% maximum bypass gives 99.900025% assembly efficiency. That allocation is a design target, not a supplier claim.

For perspective, ideal radial slit flow across the $\varnothing 64 \times 0.40$ mm land follows $Q_b = \pi h^3 \Delta \frac{p}{6\mu \ln\left(\frac{r_o}{r_i}\right)}$. At 18 psi differential and nominal total flow, a uniform gap of about 2.29 µm already carries 0.1% bypass. A 0.05% budget reduces that ideal full-circumference gap to about 1.82 µm. Local scratches need their own length/width model; this calculation is sensitivity, not an allowable surface defect.

REQUIRED EVIDENCE

Test	Purpose
Calibrated water particle challenge at 3 and 5 µm	Verify complete assembly efficiency, clean and loaded
Pore characterization / wet integrity	Detect changed pores, damage or a gross bypass route
Challenge after proof, surge and environmental exposure	Show retention survived manufacturing and loading
Media coupons and closed-perimeter element coupons	Separate media, seam and housing-seat failure modes

TEST METHOD SELECTION

Particle counters, sampling volumes, background counts and particle-size calibration must be matched upstream and downstream. For ideal independent particles with zero downstream detections, approximately 2995 challenged particles give a one-sided 95% bound of 0.1% penetration; real flow sampling and counter uncertainty require a more complete protocol.

ISO 16889 is a useful multipass-method reference, but an adapted 70 °F water/3 µm test must not be called compliant without a scope review. ASTM F316 addresses membrane pore characterization and explicitly does not establish retention on its own. ^[14] ^[13]

08 / FLOW BASIS AND CLEAN-SCREEN EQUATION

This section turns the required water flow into the speed of water through the filter.

Use superficial approach velocity over the exposed Ø58 face, without dividing by porosity again. Body loss is evaluated separately using the Ø15.75 bore velocity. Water properties are approximate ambient-pressure values at 70 °F; pressure dependence is omitted in this screening calculation. ^[12]

WATER AND GEOMETRY

$$\rho_w = 997.990 \text{ kg/m}^3$$

$$\mu_w = 0.001 \text{ Pa}\cdot\text{s}$$

$$d_{\text{face}} = 58 \text{ mm}$$

$$d_{\text{bore}} = 15.750 \text{ mm}$$

$$\dot{m}_n = 0.140 \text{ lb/s}$$

$$\dot{m}_s = 0.470 \text{ lb/s}$$

EXPOSED AREA AND APPROACH VELOCITY

$$A_{\text{face}} = \pi \cdot \frac{(d_{\text{face}})^2}{4} = 3.142 \cdot \frac{(58.000 \text{ mm})^2}{4} = 2642.079 \text{ mm}^2 \quad (3)$$

$$U_n = \frac{\dot{m}_n}{\rho_w \cdot A_{\text{face}}} = \frac{0.140 \frac{\text{lb}}{\text{s}}}{997.990 \frac{\text{kg}}{\text{m}^3} \cdot 2642.079 \text{ mm}^2} = 0.024 \frac{\text{m}}{\text{s}} \quad (4)$$

$$U_s = \frac{\dot{m}_s}{\rho_w \cdot A_{\text{face}}} = \frac{0.470 \frac{\text{lb}}{\text{s}}}{997.990 \frac{\text{kg}}{\text{m}^3} \cdot 2642.079 \text{ mm}^2} = 0.081 \frac{\text{m}}{\text{s}} \quad (5)$$

$$U_s = 0.081 \text{ m/s}$$

SCREEN-SPECIFIC RESISTANCE MODEL

$$\text{dp}_{\text{screen}} = C_i \cdot \rho \cdot U^2 + C_v \cdot U \cdot \mu \quad (6)$$

USE OF THE SUPPLIED EQUATIONS

Here $C_v = A_1 \frac{L}{D_a^2}$ and $C_i = A_2 \frac{L}{D_a}$. Their units are inverse meters and dimensionless respectively. The fitted approach preserves the user-supplied Eq. 2-2 in coherent SI; the force conversion constant is unity and is not gravitational acceleration. Never substitute a 5 µm filtration rating for average capillary diameter. ^[15]

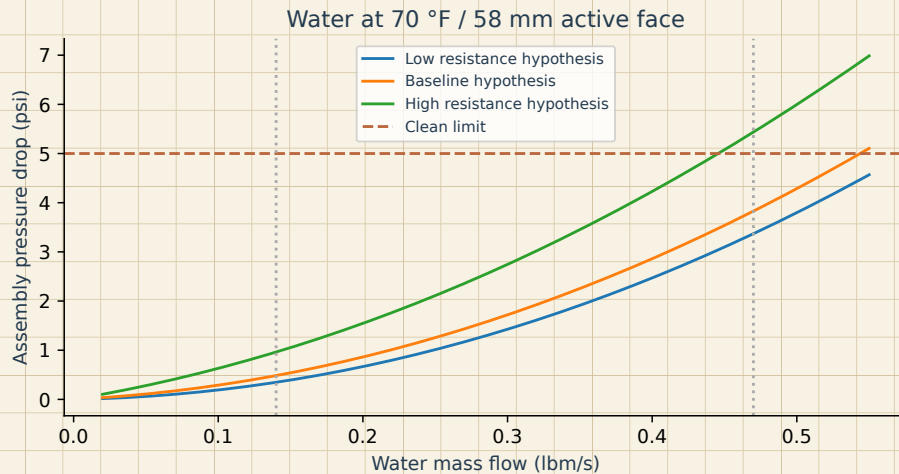
The user-transcribed 200 × 1400 example gives about 5.38 × 10⁷ /m and 116.3; those coefficients belong to that tested cloth, not to HIFLO 5-S or an arbitrary finer Dutch weave. The original NASA report file/identifier is not available here, so the transcription is not independently certified.

09 / HYDRAULIC SIZING AND SURGE MARGIN

This section estimates how much pressure the clean filter uses at normal and surge flow.

Use the same equation for each screen in series. A deliberately explicit baseline hypothesis for the fine layer is $C_v = 6 \times 10^7$ /m and $C_i = 3000$. The pair of coarse supports adds 2×10^5 /m and 40. A total stub/transition coefficient $K=3$ is applied to bore velocity. These are sensitivity inputs, not fitted supplier coefficients.

RESISTANCE ENVELOPE



HYDRAULIC HYPOTHESES

$$C_{vf} = 60000000.000 \text{ 1/m}$$

$$C_{if} = 3000$$

$$C_{vc} = 200000.000 \text{ 1/m}$$

$$C_{ic} = 40$$

$$K_{body} = 3$$

SURGE ASSEMBLY PRESSURE DROP

$$A_{bore} = \pi \cdot \frac{(d_{bore})^2}{4} = 3.142 \cdot \frac{(15.750 \text{ mm})^2}{4} = 194.828 \text{ mm}^2 \quad (7)$$

$$V_{bore} = \frac{\dot{m}}{\rho_w \cdot A_{bore}} = \frac{0.470 \frac{\text{lb}}{\text{s}}}{997.990 \frac{\text{kg}}{\text{m}^3} \cdot 194.828 \text{ mm}^2} = 1.096 \frac{\text{m}}{\text{s}} \quad (8)$$

$$DP_s = (C_{vf} + C_{vc}) \cdot \mu_w \cdot U_s + (C_{if} + C_{ic}) \cdot \rho_w \cdot (U_s)^2 + K_{body} \cdot \rho_w \cdot \frac{(V_{bore})^2}{2} \quad (9)$$

$$= (60000000.000 \frac{1}{\text{m}} + 200000.000 \frac{1}{\text{m}}) \cdot 0.001 \text{ Pa} \cdot \text{s} \cdot 0.081 \frac{\text{m}}{\text{s}} + (3000 + 40) \cdot 997.990 \frac{\text{kg}}{\text{m}^3} \cdot (0.081 \frac{\text{m}}{\text{s}})^2 + 3 \cdot 997.990 \frac{\text{kg}}{\text{m}^3} \cdot \frac{(1.096 \frac{\text{m}}{\text{s}})^2}{2} \quad (10)$$

$$= 3.829 \text{ psi} \quad (11)$$

$$DP_s = 3.829 \text{ psi}$$

WHAT THE ESTIMATE SUPPORTS

The baseline predicts 0.48 psi nominal and 3.83 psi surge, leaving 1.17 psi surge margin to the 5 psi limit. At these coefficients the fine-layer inertial term dominates surge. The smaller preliminary Ø42 disc and ½ inch stubs did not provide this margin. The inlet jet may produce nonuniform face loading; K does not model that distribution. An assembled flow test or porous-media CFD should assess effective area.

Haver publishes a pressure-drop coefficient of 9020, but its tabulated value lacks the water/velocity convention needed to fit both coefficients here. If one hypothetically interprets it as $\Delta p = 9020 \rho U^2$, then $C_i = 4510$ and the predicted surge drop is about 5.26 psi. That interpretation is not adopted as validated data; it illustrates why obtaining the actual curve is a mandatory gate. ^[1]

10 / FIT STRATEGY AND ARMOUR–CANNON CROSS-CHECK

This section explains how real mesh measurements will replace the current flow estimates.

Obtain at least five mesh-only pressure-drop points covering nominal through surge velocity at 70 °F, with a zero-flow offset check and an empty-holder subtraction. Record exposed area, pressure taps, actual thickness, alloy/heat, weave, calender condition and test-fluid properties.

FIT THE SUPPLIED MODEL

The linearized relation is $\Delta \frac{p}{\mu U} = C_v + C_i \left(\rho \frac{U}{\mu} \right)$. Use nonnegative least squares, check residuals and confidence bounds, and independently validate at another flow or sample. The delivered fit_mesh.py fits the equivalent untransformed pressure equation with scaled columns to avoid numerical conditioning issues. It requires a real CSV; this project contains no synthetic file presented as supplier data.

The transformation makes errors heteroscedastic near zero velocity. Weight by measurement uncertainty or fit the untransformed equation directly. If the tested range cannot identify both coefficients, report that limitation instead of treating an uncertain inertia term as zero.

GEOMETRY-BASED CROSS-CHECK

With matched definitions, the supplied Armour–Cannon relation is:

$$\Delta p = \frac{\tau L}{\varepsilon^2} \left(8.61 \mu a^2 U + 0.52 \rho \frac{U^2}{D_B} \right)$$

HIFLO's catalog thickness and mass give a useful consistency check: $1 - 0.95/(8000 \times 0.00018) \approx 0.340$ volumetric porosity. However, actual wire surface area per volume a , the applicable bubble-point diameter and a justified tortuosity are missing. HIFLO is not conventional Dutch twill; assigning the Dutch-weave tortuosity 1.30 to it is an unvalidated transfer. Accordingly no fabricated Armour–Cannon prediction is substituted for a measured curve. ^[11] ^[1]

The report-wide fit $f=2.49/Re+0.30$ and the straight-capillary Poiseuille model are not used for baseline prediction. The supplied research warns of aggregate correlation scatter and rejects the capillary model for predicting Dutch-twill resistance. ^[15]

SIMPLIFIED FITTING CODE

```
U = mdot / (rho * exposed_area)
X = column_stack([mu * U, rho * U**2])
scale = norm(X, axis=0)
coef, residual = nnls(X / scale, measured_mesh_dp)
Cv, Ci = coef / scale
```

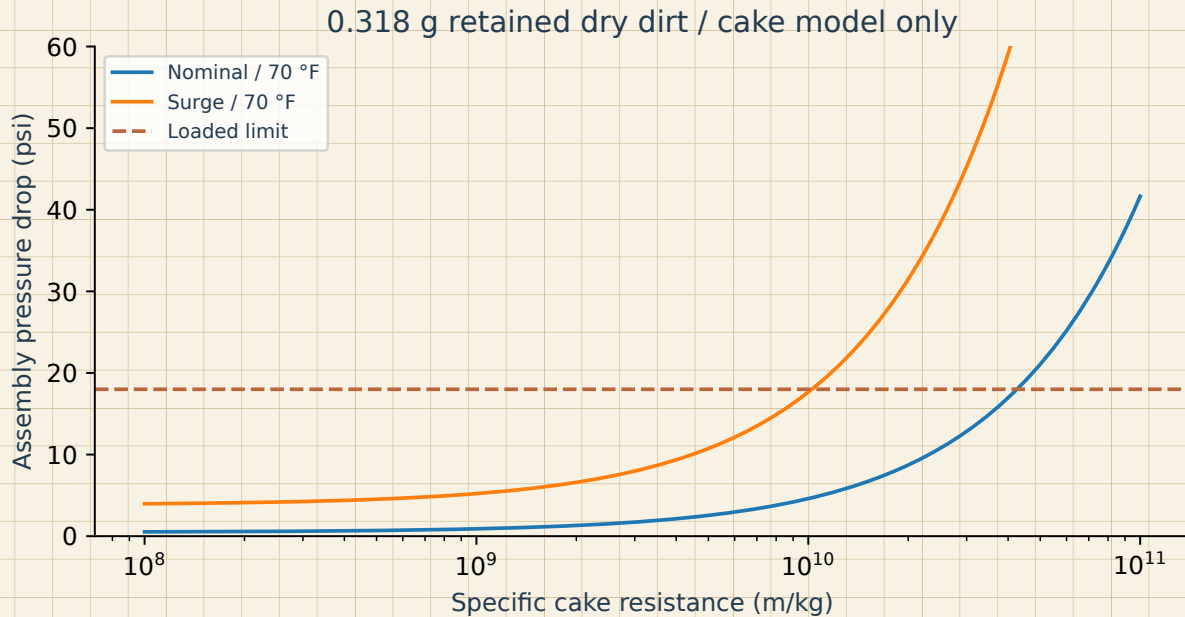
The fit coefficients belong to the tested finished material. Refit after a media change or a process that alters its hydraulic structure.

11 / RETAINED DIRT AND LOADED PRESSURE DROP

This section estimates how collected dirt makes water harder to push through the filter.

Interpret containment capacity as at least 0.318 g of retained dry particulate mass before the nominal-flow assembly drop reaches 18 psi. It is not feed mass, deposited wet mass or geometric void volume. The contaminant is still unspecified, so this requirement remains unverified.

CAKE RESISTANCE SENSITIVITY



No blockage penalty is added to the same cake; 70 °F water, 0.318 g retained dirt.

USE THE SUPPLIED SERIES-CAKE EQUATION

$$\Delta p_l = C_v \mu U + C_i \rho U^2 + \mu U \alpha_c \frac{m_d}{A_f}$$

At $\alpha = 10^9$ m/kg, calculated assembly drop is 0.90 psi nominal and 5.21 psi surge. The 18 psi nominal limit permits α up to approximately 4.26×10^{10} m/kg under the baseline clean assumptions. That is a required contaminant-performance envelope, not a measured dirt capacity. At $\alpha = 10^{11}$ m/kg the nominal loaded drop is over 40 psi, so the same mass fails.

The alternative blocked-area model replaces U by U/F . It reaches 18 psi nominal at about 87.5% completely blocked face under the baseline coefficients. That percentage cannot be converted into grams without a deposit model. Do not add blockage and cake penalties for the same dirt. Feed concentration and size-dependent capture are also required to convert retained mass into service life.

12 / HOUSING PRESSURE SCREENING

This section checks whether the housing can hold the required water pressure.

The equal-pressure housing case applies water pressure to both cavities with atmospheric pressure outside. It includes closed-end thrust transferred through the integral tube stubs. The 18 psi element drop is a separate small imbalance; 650/1300/2600 psi are not applied across the screen. No fatigue life, weld flaw tolerance or external piping loads are established.

PRESSURE AND WELD-RIM SECTION

$$p_{\text{meop}} = 650 \text{ psi}$$

$$p_{\text{proof}} = 1300 \text{ psi}$$

$$p_{\text{burst}} = 2600 \text{ psi}$$

$$r_i = 38.500 \text{ mm}$$

$$r_o = 41 \text{ mm}$$

$$S_y = 275 \text{ MPa}$$

$$S_u = 345 \text{ MPa}$$

CLOSED-END THICK-CYLINDER CHECK AT THE WELD RIM

$$\sigma_h = p_{\text{burst}} \cdot \frac{(r_o)^2 + (r_i)^2}{(r_o)^2 - (r_i)^2} \quad (12)$$

$$= 2600 \text{ pound}_{\text{force}} \text{ per square inch} \cdot \frac{(41.000 \text{ mm})^2 + (38.500 \text{ mm})^2}{(41.000 \text{ mm})^2 - (38.500 \text{ mm})^2} \quad (13)$$

$$= 41380.881 \text{ psi} \quad (14)$$

$$\sigma_z = p_{\text{burst}} \cdot \frac{(r_i)^2}{(r_o)^2 - (r_i)^2} = 2600 \text{ pound}_{\text{force}} \text{ per square inch} \cdot \frac{(38.500 \text{ mm})^2}{(41.000 \text{ mm})^2 - (38.500 \text{ mm})^2} = 19390.440 \text{ psi} \quad (15)$$

$$\sigma_r = -2600 \text{ pound}_{\text{force}} \text{ per square inch} \quad (16)$$

$$\sigma_{vm} = \left(\frac{(\sigma_h - \sigma_z)^2 + (\sigma_z - \sigma_r)^2 + (\sigma_r - \sigma_h)^2}{2} \right)^{0.5} \quad (17)$$

$$= \left(\frac{\left(\left(41380.88050314465 \text{ pound}_{\text{force}} \text{ per square inch} - 19390.440251572327 \text{ pound}_{\text{force}} \text{ per square inch} \right)^2 + \left(19390.440251572327 \text{ pound}_{\text{force}} \text{ per square inch} - (-2600 \text{ pound}_{\text{force}} \text{ per square inch}) \right)^2 + \left((-2600 \text{ pound}_{\text{force}} \text{ per square inch}) - 41380.88050314465 \text{ pound}_{\text{force}} \text{ per square inch} \right)^2 \right)}{2} \right)^{0.5} \quad (18)$$

$$= 262.611 \text{ MPa} \quad (19)$$

$$\sigma_{vm} = 262.611 \text{ MPa}$$

INTERPRETATION

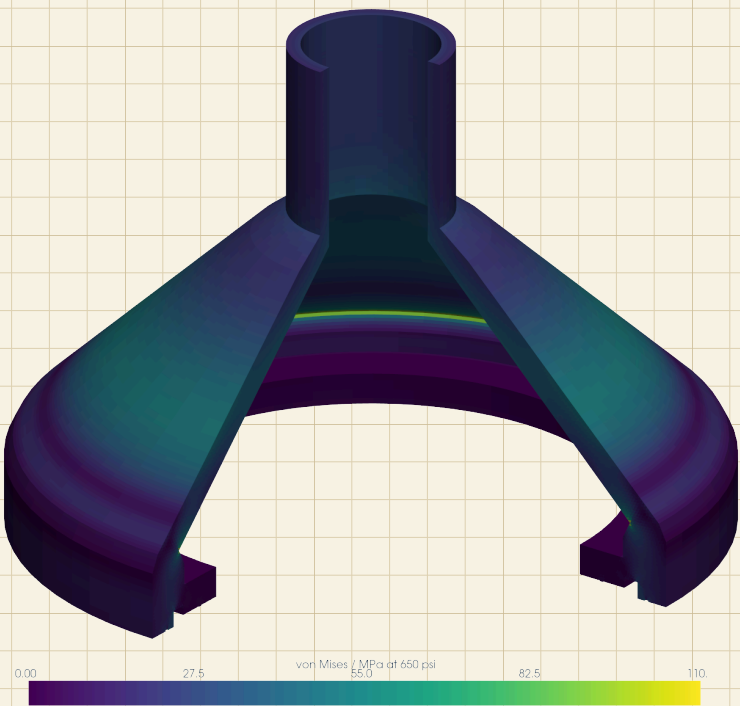
This treats the 2.50 mm fusion wall as fully effective Grade 2 material and checks a local cylindrical section. It does not represent the conical transition, weld residual stress, incomplete penetration or a burst failure model. A 10% loss of effective fusion thickness raises membrane stress by roughly 11%; the minimum weld wall must be measured on development sections.

Use a hydrostatic proof criterion of no leakage and no unacceptable permanent deformation, followed by internal integrity checks. Burst acceptance requires survival to at least 2600 psig without pressure-boundary rupture under a defined ramp. The qualification plan must define hold times, ramp rates, cycles and destructive-sample count; they are not specified by the supplied requirements.

13 / AXISYMMETRIC HYDROSTATIC SIMULATION

This section uses a computer model to estimate housing stress and movement under pressure. A custom linear-elastic axisymmetric finite-element solver uses three-node triangles with three integration points and full hoop strain N/r. It meshes the same radial profile used by Build123d. Titanium is homogeneous and isotropic. The welded rim and stop plane receive axial symmetry constraints. Pressure is applied to exposed internal faces; the seal contact track is omitted from fluid traction. There is no explicit gasket/contact preload.

650 PSIG / TRANSPARENT PRESSURE-STRESS RENDERING



Upper half only, 270-degree sweep. Undeformed geometry. MPa scale; PyVista/VTK render of the supplied ParaView-readable VTU field.

MESH SENSITIVITY / 650 PSI

Target h (mm)	Triangles	Peak VM (MPa)	Seat opening (μm)
1.2	529	64.1	3.72
0.7	1384	86.1	3.46
0.4	3914	99.7	3.40

VALIDATION AND LIMITS

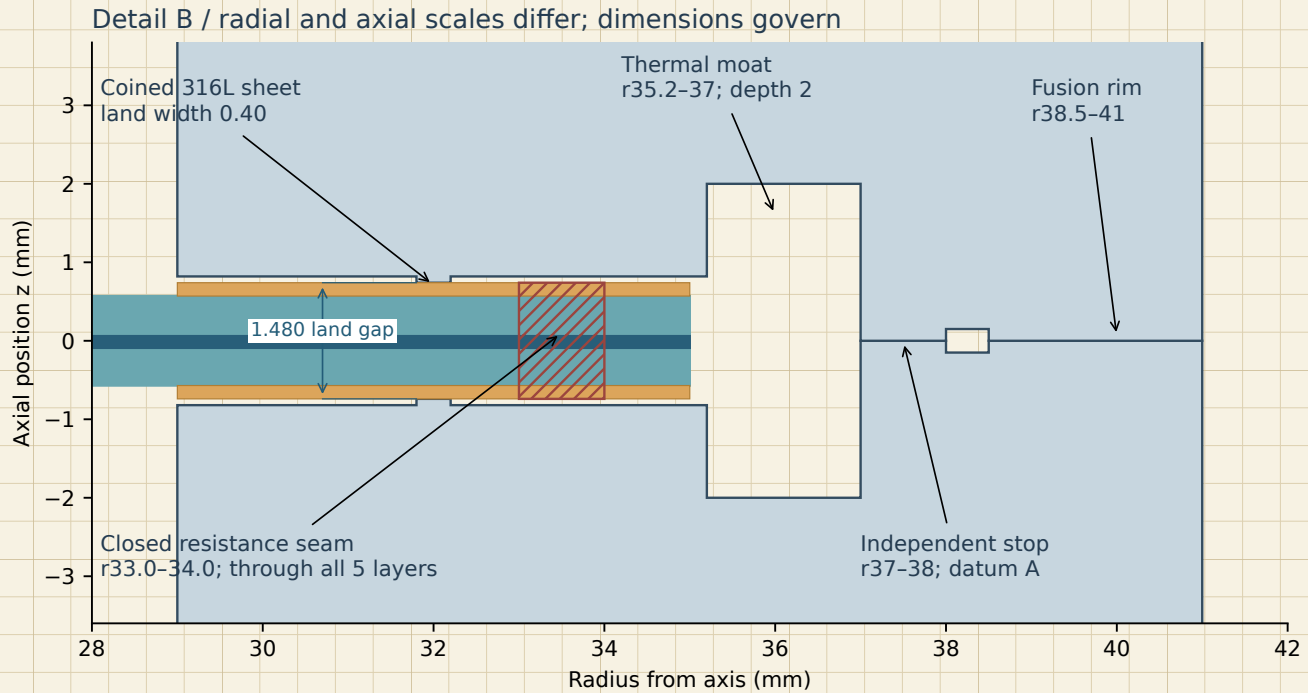
A separate closed-end thick-cylinder benchmark agrees with the Lamé stress field to 2.30% in an interior region. The finest mesh balances axial load and reaction to numerical precision. Smooth cone stress changes about 0.6% between the last two meshes. Sharp-corner peak stress rises with refinement and is not converged. The finest elastic peak is 99.7 MPa at MEOP, 199.4 MPa at proof, and 398.7 MPa by linear extrapolation to burst. The burst extrapolation exceeds minimum tensile strength locally and is not a prediction of physical rupture. Resolve fillets, contact, weld geometry and elastoplastic response before claiming pressure compliance.

14 / CRUSH SEAL AND TOLERANCE DETAIL

This section shows where the housing squeezes the metal frames to stop water going around the mesh.

Housing lands bear on solid annular sheet surfaces at Ø64. The resistance seam is outside that contact track at Ø66–68. The outermost millimeter of frame radius remains between the seam and trimmed Ø70 edge. The contact track must stay free of weld depressions, scratches and laser debris.

DETAIL B / FRAME CONTACT, INDEPENDENT STOP, WELD RIM



CAD resolves the radial placement and separate stops. Axial scale is enlarged; dimensions govern.

PROVISIONAL MANUFACTURING CONTROLS

Feature	Nominal / tolerance	Purpose
Finished element T at seal track	1.540 ±0.006 mm	Gauged AFTER resistance welding
Selective assembly gate	$T - h_A - h_B = 0.060-0.065$ mm	Measured pair accepted before final closure
Each land height from datum A	0.740 ±0.003 mm	Two halves give 1.480 ±0.006 gap
Combined face form contribution	±0.004 mm total allowance	Flatness / parallelism budget
Land mean diameter / width	Ø64 / 0.400 ±0.030 mm	Solid-frame contact region
Frame ID / final OD	Ø58 ±0.10 / Ø70 ±0.05 mm	Active area / edge capture
Pocket diameter	Ø70.40 +0.05/-0.00 mm	0.175 mm minimum radial clearance
Independent stop / weld relief	r37-38 / r38-38.5 × 0.15 deep	Stop outside element; root isolation
Surface targets	Land Ra ≤0.4 µm; no radial scratches	Provisional finish; test must validate

15 / TOLERANCE STACK AND RESIDUAL CONTACT

This section checks how small size differences affect the squeeze on the seal.

Define geometric closure before pressurization as $\delta = T - h_A - h_B + e_f + s_w - r$. Here e_f is combined form error, s_w is net axial weld closure and r represents settlement/recovery. The nominal pre-weld overlap is 60 μm . This is not the elastic compression remaining after coining.

WORST-CASE ARITHMETIC / MICROMETERS

Contribution	Range	Basis
Nominal T – hA – hB	60	1.540 – 0.740 – 0.740 mm
Element thickness deviation	–6 to +6	Finished annulus measured, not raw cloth sum
Two land heights	–6 to +6	±3 μm per half
Combined face form	–4 to +4	Explicit provisional budget
Weld closure	0 to +12	Assumed development envelope
Settlement / recovery	–8 to 0	Assumed development envelope
MEOP opening	–3.40	Equal-pressure elastic model; no preload
Residual geometric overlap	32.6 to 84.6	Worst-case sum; target 40–90 not guaranteed

CRUSH IS NOT PROOF OF PRELOAD

A coined metal washer loses most of its plastic deformation as usable elastic travel. It can retain positive geometric overlap yet lose contact force after housing distortion. The unloaded-to-MEOP model predicts about 3.40 μm increase in land gap; proof doubles that elastic estimate. The completed assembly therefore needs a nonlinear contact/coining model correlated with measured load–closure curves, springback and pressure-driven opening. The finite-element opening is only a sensitivity term because the present solver omits preload and stop contact separation.

A 0.40 mm land has about 80.4 mm² projected annular contact area per face. An assumed 250–450 MPa local coining pressure corresponds to 20–36 kN axial fixture force (not twice that force for two faces). These pressures can plastically deform Grade 2 lands as well as 316L sheets; they are not a released assembly force. Develop sheet hardness, land shape and closure together. Control on closure, force signature and coupon evidence, not force alone.

HOW TO CLOSE THE TOLERANCE REQUIREMENT

Adopt a selective assembly gate: measured T – hA – hB must be 60–65 μm before applying the other allowances. With the same form, weld and recovery bounds, residual geometric overlap is 44.6–77.6 μm at MEOP and 41.2–74.2 μm at proof. Both fit the provisional 40–90 μm geometric window. This closes the arithmetic only if the assumed process bounds hold; it does not establish residual contact pressure.

Gauge the finished perimeter at multiple azimuths and clock the halves to measured face maps. Match measured element thickness to finished land heights; reject or rematch combinations outside 60–65 μm . Measure weld shrinkage and residual preload on representative coupons. Repeat internal particle/bypass verification after proof and relevant thermal/vibration exposure. An external helium test establishes housing leakage only; it does not prove this seal or the fine-mesh perimeter.

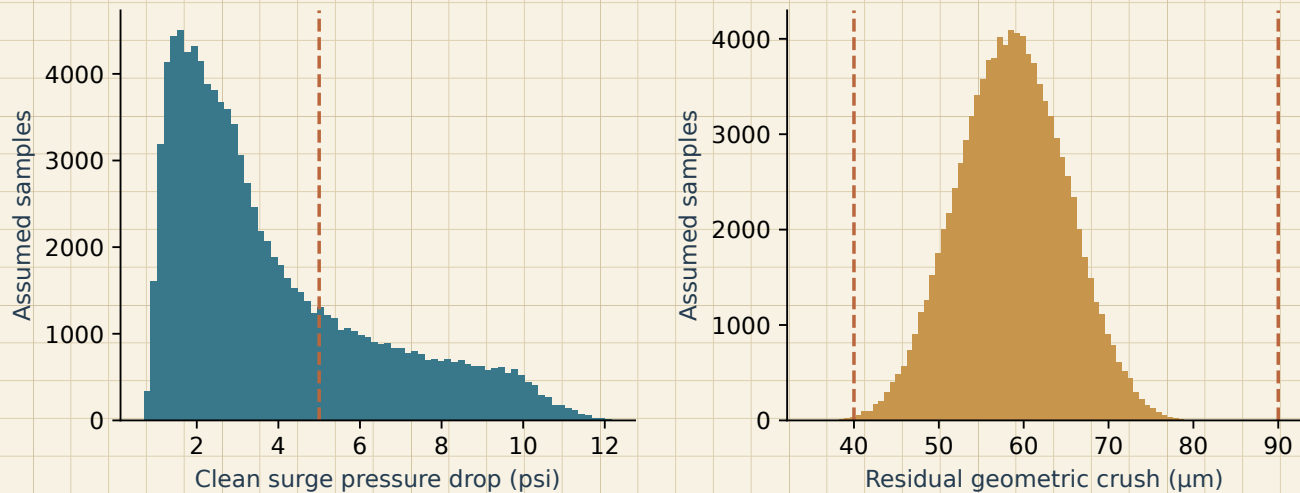
16 / MONTE CARLO / WHAT THE UNKNOWN'S CHANGE

This section tries many possible input values to show which unknowns matter most.

100,000 seeded independent draws explore assumed ranges, not measured production distributions. Cv is log-uniform from 2×10^7 to 2×10^8 / m; Ci from 300 to 10,000; cake resistance from 10^8 to 10^{11} m/kg. K is uniform 1.5–6; active diameter 57.9–58.1 mm. Temperature is fixed at the required 70 °F. Tolerance terms use the bounded ranges on the previous page, including the calculated MEOP opening.

HYDRAULIC AND GEOMETRIC SENSITIVITY

100,000 assumed draws / sensitivity, not reliability



Seed 6501300. Independent analyst assumptions; these fractions are not probabilities of field success.

FRACTION OF ASSUMED SAMPLES MEETING A SCREEN

Screen	Fraction	Interpretation
Clean nominal ≤ 5 psi	100.00%	Nominal flow is relatively forgiving
Clean surge ≤ 5 psi	73.08%	Fine-cloth inertia/resistance is critical
Loaded nominal ≤ 18 psi	87.70%	Depends strongly on contaminant α
Both hydraulic limits	64.17%	No supplier or assembly evidence included
Geometric overlap 40–90 μm	99.928%	Does not assess leakage or contact pressure
Selective pairing yield	36.78%	Randomly paired candidates within 60–65 μm

DECISION USE

Retention, actual weld defects, land damage and fatigue are not sampled. The priors are arbitrary engineering exploration bounds; different bounds and correlations change the fractions. A high geometric fraction cannot override the failing unsorted stack or prove a sealed joint. The 60–65 μm selective assembly gate removes the dimensional outliers under these assumed process bounds; it does not remove contact-model uncertainty. The next most valuable information is a $\beta 3$ -qualified fine cloth with a measured water curve, followed by seal/coining coupons. Simulation effort should narrow those uncertainties before expanding model complexity.

17 / ELEMENT SUPPORT AND DIRT LOADING MECHANICS

This section checks how the coarse mesh supports the fine mesh when water pushes on it.

At the 18 psi loaded-flow limit, the Ø58 active region carries about 328 N net axial load. One downstream coarse cloth is the primary support; the upstream cloth protects the fine layer and can react reverse loading. Direct mesh contact minimizes unsupported fine-cloth spans, but does not turn three woven layers into a bonded plate.

SCREENING THE SUPPORT LOAD PATH

For 40 mesh/in and 0.254 mm wire, pitch is 0.635 mm. The two orthogonal wire families have an equivalent metal line area $t_e = 2\frac{\pi d_w^2}{s} \approx 0.160$ mm. Using 170 MPa only as a reference solid-sheet yield stress gives about 27.1 N/mm line tension. A full-circumference upper-bound axial capacity with $\sin(\text{slope})=1$ is approximately 4.94 kN. This is an optimistic strength bound, not a support qualification: real slopes, anisotropy, wire crimp, boundary capture and local fine-wire bending govern. ^[8]

A taut-membrane scale estimate using $w \approx \left(3p \frac{a^4}{8Et_e}\right)^{\frac{1}{3}}$ gives roughly 1–2 mm deflection at 18 psi for effective modulus 193–30 GPa. That is consistent with needing millimeter-scale cavity clearance, not an assumption of a rigid flat disc. The opening-side cavity has 4.5 mm nominal shoulder height. This estimate is not a predictive woven-cloth constitutive model, and the existing envelope CAD is not a structural mesh model.

VERIFICATION NEEDED

Pressure-cycle a completed element to at least the required operating differential, including the measured dirty surge transient; inspect permanent set, shedding and retention. If support fails, a housing-integral support pattern can preserve the seven-piece count, but it reduces accessible area and changes hydraulics. No such backing grid is included or credited in this baseline.

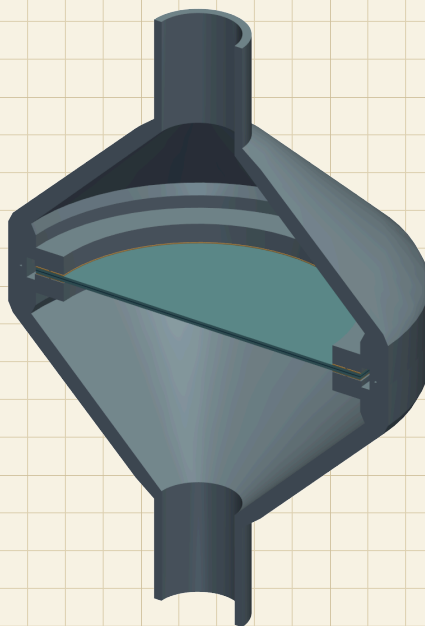
For the nominal 1 mm resistance-seam band at mean radius 33.5 mm, average axial shear from 328 N is about 1.56 MPa. That low average does not prove closed pores, peel resistance, uniform nugget overlap or resistance to wire pullout. Fine-layer edge bypass remains a functional-test requirement.

18 / ORBITAL WELD, THERMAL SEPARATION AND FIT-UP

This section explains how to weld the housing and protect the filter from weld heat.

The element plane and orbital seam are both centered at $z=0$ because the cups are identical and opposed. There is zero axial separation. The nearest nominal fusion root is radius 38.5 mm; the frame OD is radius 35 mm, giving 3.5 mm radial separation. An annular moat at radius 35.2–37 mm, 2 mm deep per half, interrupts direct conduction near the frame. This is an explicit development departure from the research's requested axial thermal offset.

HOUSING AND PACK / CUTAWAY ISOMETRIC



Rendered CAD cutaway. Radial thermal relief is visible outside the captured element.

WHY RADIAL SPACING IS NOT YET SUFFICIENT EVIDENCE

Using titanium $k \approx 21.8 \text{ W/(m K)}$, density 4510 kg/m^3 and assumed heat capacity 520 J/(kg K) , the diffusion length $\sqrt{\alpha t}$ is about 9.6 mm over 10 s, already larger than the radial clearance. This scale estimate is not a transient temperature prediction. Weld heat input, travel speed, fixture heat sinking, radiation and contact conductance are missing. ^[6]

Use full weld/HAZ inert shielding, including the internal purge and trailing protection. Monitor purge quality and oxygen pickup through qualified process controls. Keep weld heat, crater closure and start/stop overlap away from a presumed single acceptable temperature criterion; establish acceptance using instrumented coupons, metallography and mechanical/cleanliness checks. ^[10]

Provisional fit-up targets: weld-end squareness within 0.03 mm, radial mismatch $\leq 0.05 \text{ mm}$, and root gap 0–0.05 mm while the hard stop is engaged. These are development targets requiring WPS confirmation. Provide accessible fixturing faces without clamping on the fragile element.

19 / MANUFACTURING PROCESS PLAN

This section gives the steps for making, assembling and inspecting the filter.

Make the element in a clean stainless fabrication stream, then assemble it into separately prepared titanium housings. Precut the washer ID before placing it on the mesh. Never laser-cut the central opening through the assembled five-layer pack.

TRAVELER / DEVELOPMENT SEQUENCE

Step	Operation	Record / acceptance evidence
10	Receive certified alloy stock and cloth	Heat/lot, weave, origin, thickness and material certificate
20	Machine one housing design twice	Datum A stop; bore axis B; seat/land map; minimum wall
30	Prepare annular 316L blanks	Ø58 ID first; stock OD for later trim; deburr/clean
40	Calender and stack coarse/fine/coarse	0.480 mm coarse target; fine weave remains intact
50	Resistance-weld closed annulus	Through all five layers; schedule, current, force, overlap
60	Trim only final outer outline	Ø70 OD outside closed band; ≥1 mm nominal edge capture
70	Clean and inspect element	Debris, recast, loose wires, pore integrity and flow
80	Gauge finished seal track	T=1.540±0.006 mm AFTER seam; inspect multiple azimuths
90	Select pair; fixture closure / coin	60–65 µm measured overlap; force–closure and face maps
100	Orbital-weld housing to housing	Qualified full-penetration titanium seam and purge
110	Final clean and pressure verification	Proof, external leak, internal retention and shedding
120	Mark and package	Traceable serial/lot; protected weld stubs; cleanliness record

PARAMETERS THAT REMAIN EXPERIMENTAL

Electrode material/shape, number of passes, nugget pitch, welding current, force and time need development on the actual five-layer stack. Use a continuous seam or qualified overlapping nuggets that leave no circumferential bypass route. Scattered retention spots or separate frame-to-coarse welds are insufficient evidence. A resistance-welded annulus can still contain connected pore paths; sectioning and functional challenge are required.

Do not translate solid-sheet weld settings directly to woven cloth. Inspect after element fabrication and again after housing welding. Passivation/cleaning of stainless must be compatible with the fine media; cleaning of the hybrid final assembly must also be qualified for titanium.

20 / QUALIFICATION AND RELEASE GATES

This section lists the tests and evidence still needed before the filter can be used.

The design is ready for supplier discussion, tooling feasibility and coupon development. It is not ready for fabrication as a flight/service-qualified filter. No test certificates, supplier quotations or accepted weld procedures are included.

EVIDENCE REQUIRED BEFORE RELEASE

Gate	Pass evidence	Current disposition
G1 / media selection	$\beta_3 \geq 1000$ and defined 5 μm retention; actual Cv/Ci	OPEN; possible finer media needed
G2 / closed element perimeter	Microscopy/sections plus bypass and shedding tests	OPEN; five-layer seam unqualified
G3 / frame/housing sealing	Contact/coining validation; pressure and exposure tests	OPEN; dimensions alone insufficient
G4 / housing pressure boundary	WPS/PQR, NDE, proof and burst qualification	OPEN; elastic screening only
G5 / flow and capacity	≤ 5 psi clean and ≥ 0.318 g at ≤ 18 psi	OPEN; dust and real media curve missing
G6 / mass and installation	≤ 1 lb finished; compatible line alloy and weld access	CALCULATED mass; fit-up to confirm

RECOMMENDED TEST SEQUENCE

First characterize media coupons and full closed-seam elements. Next develop coining and housing-weld coupons with instrumented temperature and closure measurements. On completed assemblies, establish clean flow and particle retention before proof, then repeat internal integrity after proof. Run nominal and surge flow during controlled dirt loading; record retained dry mass, pressure drop, particle penetration and shedding. Use separate qualified samples for destructive burst so a destroyed item is never mistaken for a deliverable.

Select sample counts and confidence targets with the qualification authority. Hold times, leakage acceptance, allowable permanent set and environmental spectra require explicit approval in the eventual test procedure. External helium leakage and internal filtration integrity are separate acceptance lines.

If axial element-to-fusion offset becomes mandatory, revise the identical-half architecture rather than relabel radial clearance as axial separation.

If stainless installation lines are mandatory, revisit material selection or allocate a qualified transition outside the seven-part filter definition.

21 / REPRODUCIBILITY AND SIMPLIFIED SIMULATION CODE

This section explains how to rerun the code and recreate the drawings and calculations.

All project code and generated outputs reside under filter/. Kip source code is unchanged. The local uv environment is locked; build123d is the requested CAD library. Full calculations, seeds, meshes, pressure fields, STEP files and rendered views are included.

BUILD SEQUENCE

```
uv sync
uv run python cad_model.py
uv run python hydrostatic_fem.py
uv run python analysis.py
uv run python figures.py
uv run python author_document.py
uv run kip check
uv run kip build --output F100_engineering_design.pdf
```

The requirements check intentionally leaves untested service requirements unverified. A successful document build is not engineering qualification.

SIMPLIFIED AXISYMMETRIC FINITE-ELEMENT ASSEMBLY

= Strains: radial, axial, hoop, engineering shear.

```
B[0, 0::2] = dN_dr
B[1, 1::2] = dN_dz
B[2, 0::2] = N / radius
B[3, 0::2] = dN_dz
B[3, 1::2] = dN_dr
Ke += B.T @ D @ B * (2*pi*radius*area/3)
K = assemble_sparse(element_matrices)
u[free] = spsolve(K[free][:,free], load[free])
stress = D @ B @ element_displacement
```

Full code includes three-point area integration, consistent pressure traction, closed-end thrust, boundary conditions, reaction balance, a Lamé benchmark and three mesh densities.

PARAVIEW DELIVERABLE

Open output/analysis/housing_pressure_650psi.vtu in ParaView and color by cell von_Mises_MPa. The delivered transparent PNG was produced with PyVista/VTK, not a running ParaView application. paraview_render.py reproduces a true ParaView transparent screenshot with pypython when that application is installed. No background, ground plane or decorative geometry is included in the stress render.

The .vtu file is a swept axisymmetric result, not an independent three-dimensional solution. Housing stress does not include filter preload or model the woven cloth; the field must not be reused as evidence for element strength or sealing.

22 / AUDITABLE CALCULATIONS AND VERIFICATION RECORD

This section shows how the filter weight was calculated and checked.

The mass estimate uses actual housing CAD volume, solid blank-frame mass and catalog areal weights for the mesh envelopes. Solid-disc envelope volumes are not used as mesh metal mass. Coarse mass is based on the raw construction; calendering does not remove metal.

CAD VOLUME AND AREAL MASSES

$$V_{\text{half}} = 38331.113 \text{ mm}^3$$

$$\rho_{\text{ti}} = 4510.000 \text{ kg/m}^3$$

$$m_{\text{frames}} = 0.004 \text{ kg}$$

$$m_{\text{coarse}} = 0.011 \text{ kg}$$

$$m_{\text{fine}} = 0.004 \text{ kg}$$

$$\text{growth} = 1.200$$

DRY FINISHED MASS WITH A DEVELOPMENT RESERVE

$$m_{\text{dry}} = 2 \cdot V_{\text{half}} \cdot \rho_{\text{ti}} + m_{\text{frames}} + m_{\text{coarse}} + m_{\text{fine}} = 2 \cdot 38331.113 \text{ mm}^3 \cdot 4510.000 \frac{\text{kg}}{\text{m}^3} + 0.004 \text{ kg} + 0.011 \text{ kg} + 0.004 \text{ kg} = 0.802 \text{ lb} \quad (20)$$

$$m_{\text{reserve}} = \text{growth} \cdot m_{\text{dry}} = 1.200 \cdot 0.802 \text{ lb} = 0.962 \text{ lb} \quad (21)$$

$$\text{MS}_{\text{mass}} = \frac{1 \cdot \text{lb}}{m_{\text{reserve}}} - 1 = \frac{1 \cdot \text{lb}}{0.962 \text{ lb}} - 1 = 0.039 \quad (22)$$

$$\text{MS}_{\text{mass}} = 0.039$$

MASS SCOPE

This is dry hardware mass including both integral stubs, before any additional mounting hardware, line transitions or external piping. The 20% multiplier is a design reserve, not a Monte Carlo tolerance prediction. Trapped water mass is excluded because the requirement has been interpreted as dry component mass. The estimate leaves only about 0.038 lb after that reserve; a substantial new support or transition must trigger reweighing.

23 / VERIFICATION LEDGER

This section separates requirements already checked from those still waiting for evidence.

Only mass by analysis and the seven-component count have supporting verification entries. The remaining service requirements deliberately stay OPEN until the evidence identified in the release gates exists. This ledger distinguishes successful calculation execution from qualification of manufactured hardware.

KIP REQUIREMENT LEDGER

ID	Requirement	Method	Evidence	Result	Status
R01	The fully built filter shall contain 650 psig water at 70 F.	Test	-	-	OPEN
R02	The fully built filter shall withstand 1300 psig hydrostatic proof.	Test	-	-	OPEN
R03	The fully built filter shall have burst pressure at least 2600 psig.	Test	-	-	OPEN
R04	The filter shall provide the specified 5 micrometer filtration rating.	Test	-	-	OPEN
R05	The completed assembly shall retain 99.9 percent at 3 micrometers.	Test	-	-	OPEN
R06	The assembly shall pass 0.14 lbm/s nominal water flow.	Test	-	-	OPEN
R07	The assembly shall pass 0.47 lbm/s surge water flow.	Test	-	-	OPEN
R08	Clean assembly pressure drop shall be at most 5 psi.	Test	-	-	OPEN
R09	The assembly shall retain at least 0.318 g dry contaminant.	Test	-	-	OPEN
R10	Loaded assembly pressure drop at capacity shall be at most 18 psi.	Test	-	-	OPEN
R11	Dry assembly mass including integral stubs shall be at most 1 lb.	Analysis	mass_calc	$0.962 \text{ lb} \leq 1 * \text{lb}$	PASS
R12	Service water temperature shall be 70 F.	Test	-	-	OPEN
R13	The baseline assembly shall contain seven components.	Inspection	Seven valid noninterfering CAD solids; cad_model.py	$7 = 7$	PASS

24 / REFERENCES AND SOURCE LIMITATIONS

This section lists the sources used and explains what they do and do not prove.

Supplier references establish specific catalog facts and sourcing leads, not quotations or certificates for delivered material. All online sources were checked on 2026-09-14. Analysis inputs not attributed to a supplier are explicitly engineering hypotheses. The user-supplied research is retained as the authority for the architecture and selected hydraulic equations.

TRACEABLE REFERENCES

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